Prediction Bottlenecks Don't Discover Causal Structure (But Here's What They Actually Do)

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Submitted to NeurIPS 2026. Do not distribute.

Abstract

A Mamba state-space model trained only for next-step prediction appears to recover Granger-causal structure through a strikingly simple readout: $\mathbf{S} = |\mathbf{W}_{\text{out}}\mathbf{W}_{\text{in}}|$. Early experiments suggested this generalized across architectures and benefited from interventional data with $p < 10^{-5}$. We walk this claim through a five-stage falsification pipeline. The method-level claim does not survive: (i) a plain linear bottleneck does as well or better, (ii) tuned Lasso beats it on synthetic and real data (NOAA climate, ETFs, Lorenz-96), and (iii) the headline intervention advantage is $\sim$60% a size-matched-data confound and $\sim$40% a target-corruption artifact specific to a non-standard random-forcing scheme. What survives is a narrow characterization result. We hope the falsification scaffold is as useful to the community as the positive findings we could not salvage.

1 Introduction

A natural hope in time-series modelling is that a forecaster trained only on next-step prediction will, as a byproduct, expose causal or dependency structure through its learned weights *causal discovery as a byproduct of prediction*. If true, every pretrained forecaster would double as a causal-discovery engine.

This paper reports the result of taking that hope seriously. Our starting point was the observation that for a Mamba selective state-space model (Gu and Dao, 2023) trained for next-step MSE, the product $\mathbf{S} = |\mathbf{W}_{\text{out}}\mathbf{W}_{\text{in}}|$ recovers Granger-causal graphs with striking reliability on synthetic VAR benchmarks. Initial experiments suggested the phenomenon generalized across architectures, emerged from training rather than initialization, and most provocatively that interventional data amplified the bottleneck's advantage with $p < 10^{-5}$ across twelve (K, T) configurations.

Each of these claims turned out to be less robust than it first appeared. In the spirit of ICBINB, our contribution is *negative*: we show that a natural method candidate fails in five subtly different ways, identify the controls that caused the failures, and report the narrow findings that remain valid.

Contributions.

1. We falsify architecture specificity: a linear bottleneck matches or beats the SSM under matched capacity and seeds.
2. We falsify competitiveness with sparse regression: tuned Lasso beats the bottleneck on synthetic CauseMe-style benchmarks and on real observational data (NOAA climate, sector ETFs, Lorenz-96).
3. We falsify the headline intervention advantage: under a size-matched control and proper do$(X = c)$ interventions, the effect collapses; the residual reproduces (and is larger) under classical Granger.
4. We document what survives mild-nonlinearity, sample-efficiency, target-corruption robustness as characterization, not method, claims, and release a reusable falsification scaffold.

Table 1: Bottleneck vs. best tuned baseline on the focused stress grid.

Metric vs. {OLS, Ridge, Lasso, RRR}	Mean AUROC delta	Win rate
Bottleneck flat-lag AUROC	-0.050	0.12
Bottleneck prediction MSE	$+0.286$	0.00
Best baseline by AUROC: Lasso (38/48), RRR (10/48)		
Best baseline by MSE: Lasso (48/48)		

Table 2: AUROC on three real/benchmark datasets. The bottleneck tops only climate, where ground truth is weak (3 edges, K=6); on the clean Lorenz-96 benchmark, classical causal methods lead by a wide margin.

Method	Climate (NOAA) K=6, T=757	Finance (ETFs) K=10, T=1893	Lorenz-96 K=10, T=1500
Weight-Proj (SSM)	**0.864**	0.438	0.802
PCMCI	0.704	0.510	**0.986**
Tuned Lasso	0.630	0.500	0.974
Lagged Bottleneck	0.617	0.514	0.920
Tuned Ridge	0.617	**0.696**	0.974
Bivariate Granger	0.605	0.498	0.979

2 Method: The Extraction We Are Falsifying

For a K-variable series $\mathbf{X} \in \mathbb{R}^{T\times K}$, we train a model with input projection $\mathbf{W}_{\text{in}} \in \mathbb{R}^{d\times K}$ and output projection $\mathbf{W}_{\text{out}} \in \mathbb{R}^{K\times d}$ on next-step MSE. No causal-specific loss. After training, we extract $\mathbf{S} = |\mathbf{W}_{\text{out}}\mathbf{W}_{\text{in}}| \in \mathbb{R}^{K\times K}$ with $\mathbf{S}_{i,j}$ interpreted as the strength of $j \to i$, zero the diagonal, normalize, and threshold. For architectures with explicit lagged input (the lagged bottleneck variant), we extend to $\mathbf{S}^{(\tau)} = |\mathbf{W}_{\text{out}}\mathbf{W}_{\text{in}}^{(\tau)}|$ and evaluate via flat-lag AUROC against ground-truth adjacency.

3 Five Falsifications

F1. Architecture does not matter. A pre-registered sweep at matched parameter counts and seeds across four datasets and five architectures (Mamba SSM, Transformer, LSTM, MLP, Linear) finds the linear bottleneck matches or beats the SSM on every dataset (mean AUROC: VAR(1)-chain K=5: SSM 1.00, Linear 1.00; VAR(1)-random K=10: SSM 0.77, Linear **0.99**; regime-switch K=3: both 1.00; Lorenz K=3: SSM 0.52, Linear **0.94**). The "emergence" is the linear bottleneck doing low-rank lag-1 regression; the SSM achieves the same estimator at higher parameter cost.

F2. Tuned Lasso beats the bottleneck on graph recovery. Across a focused stress grid ($K \in \{10, 20\}$, $T \in \{150, 300\}$, max-lag 8, Table 1) the bottleneck wins only 12% of graph-recovery runs and 0% of MSE runs against tuned baselines. On tigramite CauseMe-style processes the gap widens with K: Lasso reaches AUROC 0.98 at K=20 versus 0.73 for the bottleneck.

F3. Real data: not competitive. We evaluate six methods on three datasets of progressively cleaner ground truth (Table 2): 63 years of NOAA monthly climate indices (ENSO, NAO, PDO, AMO, SOI, PNA) with teleconnection edges from the climate literature (Trenberth and Hurrell, 1994; Delworth and Mann, 2000; Newman et al., 2016); 7.5 years of daily SPDR sector ETF log returns (K=10, soft labels); and Lorenz-96 (K=10, F=10) with mathematically unambiguous ground truth (Tank et al., 2021; Nauta et al., 2019). We deliberately exclude the ENSO↔SOI pair from climate ground truth because SOI and the Niño 3.4 SST anomaly are the same physical signal with opposite sign, not a causal edge. On Lorenz-96, where ground truth is unambiguous, *PCMCI and bivariate Granger dominate* (0.986, 0.979); the lagged bottleneck trails at 0.920 and the SSM bottleneck is worst at 0.802. The bottleneck is not the best method on any dataset with clean ground truth.

F4. Interventions: a confound story. An earlier experiment (Exp 19) compared AUROC($\mathbf{X}_{\text{obs}}$) versus AUROC($\mathbf{X}_{\text{obs}} \cup \mathbf{X}_{\text{int}}$) for both bottleneck and Lasso, reporting 12/12 configs significant at

$p < 10^{-5}$ in favor of the bottleneck. Two controls dissolve it. *Size-match (Exp 22):* Exp 19 conflated more data with intervention content; adding a fresh observational arm of equal total size collapses the effect to ∼40% of the raw gap. *Intervention type (Exp 23):* the original "intervention" was a per-step random forcing $x_{i,t} \leftarrow s \cdot \epsilon_{i,t}$, not a standard do-intervention. Replacing it with constant clamps do$(X_i = c)$ gives only $3/12$ configs significant (mean gap $+0.002$); soft (added-noise) interventions give $6/12$, mean gap $+0.008$. Only random forcing produces a stable bottleneck gain, best explained as robustness to corrupted target rows.

F5. Method-agnosticity. The residual effect is not unique to the bottleneck. Adding PCMCI and bivariate Granger to the size-matched control: the bottleneck shows a significant size-matched gain ($+0.026$ to $+0.054$, $p < 10^{-3}$), *but so does Granger*, with larger effects at higher K ($+0.040$ at K=20, $+0.095$ at K=30). Lasso uniquely fails, consistent with per-equation regression breaking under target corruption. The intervention effect, insofar as it survives controls, is method-agnostic and matches a target-corruption-robustness story rather than causal-structure extraction.

4 What Survives

After five falsifications, three narrow positives remain. **(i) Mild-nonlinearity regime:** in a regime sweep over noise, density, and nonlinearity strength, the bottleneck outperforms Lasso at `nonlinear=0.3` (78% win rate over 3 seeds and 9 hyperparameter cells). At stronger nonlinearity, both methods fail equally. **(ii) Sample efficiency:** bottleneck gains from additional observational data exceed Lasso's by ~ 0.07 AUROC at $K \in \{20, 30\}$ under size-matched control arms modest but useful where collecting more observations is cheap. **(iii) Target-corruption robustness:** the bottleneck's shared $\mathbf{W}_{\text{out}}$ absorbs per-step random forcing better than Lasso's per-equation fits. This explains the residual intervention effect, but is a reliability result, not a causal-discovery one.

These are characterization claims. We do not propose the bottleneck as a causal discovery method.

5 Lessons

Size-matched controls beat seed counts. Exp 19 had 5 seeds. Exp 21 had 20 seeds and reported $p < 10^{-5}$ across 12 configs. Both found the same effect; neither would have flagged it as a sample-size confound. Adding seeds tightens a noisy measurement; it cannot diagnose a missing structural control.

Pre-register the intervention scheme. The gap between "random forcing" and proper do$(X_i = c)$ is the difference between a significant effect and a null, and it is not obvious from the code. We should have written down which intervention semantics counted as "interventional data" before running the experiment.

Prediction baselines are not causal baselines. Our intervention experiments compared the bottleneck only against prediction-fit baselines (OLS, Ridge, Lasso, RRR). It was natural to miss that classical causal methods (Granger, PCMCI) would benefit from the same intervention scheme by the same mechanism. Any future "neural causal discovery" claim should include at least one classical causal baseline in the intervention control arm.

6 Conclusion

We set out to find a free causal-discovery method hiding inside prediction bottlenecks and could not. The headline effects either fail under size-matched controls, are specific to a non-standard intervention scheme, or reproduce as larger effects in classical causal methods. What survives is a narrow set of characterization findings and a reusable falsification scaffold.

Code. All experiments, data loaders, and a one-line reproduction script will be released upon acceptance (anonymous URL provided to reviewers).

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